

The Customer's Segmentation in a Shopping Mall

Summary:

In this report, I've gathered the data from Kaggle titled "Customer's Segmentation" with 200 data sets of a customer's with their Annual Income and Spending Score (1-100), and started to divide them into clusters based on how much they earn in their Annual Income and How do they rate their spending things inside the shopping mall. And the clusters was formed into 5 groups which are cluster 1(Average Income and Average Spending), cluster 2 (High Income and High Spending), cluster 3 (Low Income and High Spending), cluster 4 (High Income and Low Spending), and cluster 5 (Low Income and Low Spending). And there are 81 customers in Cluster 1, 39 customers in Cluster 2, 22 customers in Cluster 3, 35 customers in Cluster 4, 23 customers in Cluster 5. And they are grouped based on their nearest centroids, which was shown in a visual graph of K-Means titled "The Customers Segmentation".

Scenario:

The shopping mall owner wants to create a marketing campaign for their customers and it started to collect Customer data which are their Annual Income and Spending Score (1-100) which can be represented as from low spending to highest spending, a higher spending score represented as customers who spends more at the mall. Aside from that the shopping mall owner wants to divide its customers into groups based on their Annual Income and Spending Score.

Objectives:

The objectives of this study is to divide the shopping mall's customers into different groups based on their Annual Income and Spending Score (1-100)

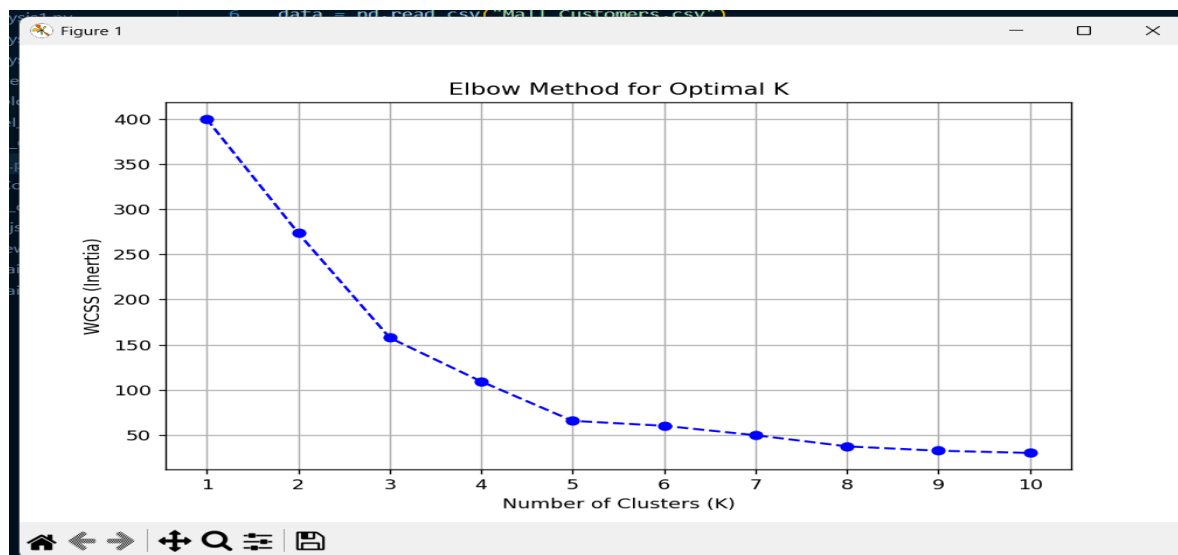
Methods:

In this report, I've gathered the data that came from kaggle and downloaded its csv file and used the shopping malls customer's Annual Income and Spending Score(1-100) and started to divide them into clusters by the use of K-means clustering as

An algorithm to divide the customer's into clusters. K-means clustering is a type of an unsupervised machine learning algorithm that is being used to group the unlabelled data into k distinct clusters, and it is used to assign the data points to their nearest cluster center or centroids until it makes the efficient for pattern recognition and segmentation.

But before dividing them into clusters, I started to use the Elbow Method to find its k distinct clusters or their number of clusters. The Elbow Method is a type of visual heuristic that is used to find the optimal number of clusters (k) in K-Means. Its calculations are highly based On the calculations of their WCSS(Within-Cluster Sum of Squares) it is a kind of model variation that plots the individual data points to find its elbow or rate of where its rate of decrease drops.

In performing the Elbow Method and K-Means I used Python as a programming language for this report, because it has libraries and frameworks that can be used for machine learning, data science, and artificial intelligence. I used the python frameworks like matplotlib, pandas, sklearn(KMeans and StandardScaler). With those frameworks it is possible to visualize and analyze those 200 data sets in performing the K-means. Aside from performing the Elbow method I started to compute the customers distance and the clusters centroids in order to find which individual customer's distance are nearest to each centroids.



In the image above it is represented Elbow Method graph that used to identify the Optimal K or their number of clusters and the image shows the x-axis(Number of Clusters (k)) and y-axis(WCSS(Inertia)) which measures how close the data points are to their assigned centroids, When the number of clusters increases then the WCSS is decreasing because it starting to form a small clusters, and the points started to change from a steep decline to gentle decline. As a result the graph suggested that the number of optimal k for this report is five clusters for segmenting the data. The reason why k=5 is chosen is because it forms an elbow shape from 400 until it drops down to 50 which it decreases rapidly and the other points have a smaller decrease to each other. If we intend to add more clusters higher than five it will not improve its clustering because the model will make it much more complex to view its data points. WCSS is the representation of the total squared distance of all the data points from their cluster centroids and it is the sum of the squared distances of every customer in the dataset.

Since were able to find its optimal K or their K distinct clusters which k=5, I clustered the mall's individual customers into 5 groups which are:

Cluster Interpretations:

- Cluster 1 - Average Income, Average Spending
- Cluster 2 - High Income, High Spending
- Cluster 3 - Low Income, High Spending
- Cluster 4 - High Income, Low Spending
- Cluster 5 - Low Income, Low Spending

These are the possible groups that can be used for clustering the mall's customers.

Results:

- From Figure 1 to 5 it is the list of customers with their ID, Annual Income and Spending Score in different Clusters(1-5).

Customers in Each Cluster

===== Cluster 1 =====

	CustomerID	Annual Income (k\$)	Spending Score (1-100)
43	44	39	61
46	47	40	55
47	48	40	47
48	49	40	42
49	50	40	42
..	...	...	...
121	122	67	40
122	123	69	58
126	127	71	35
132	133	72	34
142	143	76	40

Figure 1: Shopping Mall's Customers in Cluster 1 who earn an Average Annual Income and have an Average Spending Score.

===== Cluster 2 =====

	CustomerID	Annual Income (k\$)	Spending Score (1-100)
123	124	69	91
125	126	70	77
127	128	71	95
129	130	71	75
131	132	71	75
133	134	72	71
135	136	73	88
137	138	73	73
139	140	74	72
141	142	75	93
143	144	76	87
145	146	77	97
147	148	77	74
149	150	78	90
151	152	78	88
153	154	78	76
155	156	78	89
157	158	78	78
159	160	78	73
161	162	79	83
163	164	81	93
165	166	85	75
167	168	86	95
169	170	87	63
171	172	87	75
173	174	87	92
175	176	88	86
177	178	88	69
179	180	93	90
181	182	97	86
183	184	98	88
185	186	99	97
187	188	101	68
189	190	103	85
191	192	103	69
193	194	113	91
195	196	120	79
197	198	126	74
199	200	137	83

Figure 2: Shopping Mall's Customers in Cluster 2 who earn a High Annual Income and have a High Spending Score.

===== Cluster 3 =====			
	CustomerID	Annual Income (k\$)	Spending Score (1-100)
1	2	15	81
3	4	16	77
5	6	17	76
7	8	18	94
9	10	19	72
11	12	19	99
13	14	20	77
15	16	20	79
17	18	21	66
19	20	23	98
21	22	24	73
23	24	25	73
25	26	28	82
27	28	28	61
29	30	29	87
31	32	30	73
33	34	33	92
35	36	33	81
37	38	34	73
39	40	37	75
41	42	38	92
45	46	39	65
Number of Customers in Each Cluster			

Figure 3: Shopping Mall's Customers in Cluster 3 that has a Low Annual Income but has a High Spending Score.

===== Cluster 4 =====			
	CustomerID	Annual Income (k\$)	Spending Score (1-100)
124	125	70	29
128	129	71	11
130	131	71	9
134	135	73	5
136	137	73	7
138	139	74	10
140	141	75	5
144	145	77	12
146	147	77	36
148	149	78	22
150	151	78	17
152	153	78	20
154	155	78	16
156	157	78	1
158	159	78	1
160	161	79	35
162	163	81	5
164	165	85	26
166	167	86	20
168	169	87	27
170	171	87	13
172	173	87	10
174	175	88	13
176	177	88	15
178	179	93	14
180	181	97	32
182	183	98	15
184	185	99	39
186	187	101	24
188	189	103	17
190	191	103	23
192	193	113	8
194	195	120	16
196	197	126	28
198	199	137	18
Number of Customers in Each Cluster			

Figure 4: Shopping Mall's Customer who has a High Annual Income but has a Low spending Score.

===== Cluster 5 =====			
	CustomerID	Annual Income (k\$)	Spending Score (1-100)
0	1	15	39
2	3	16	6
4	5	17	40
6	7	18	6
8	9	19	3
10	11	19	14
12	13	20	15
14	15	20	13
16	17	21	35
18	19	23	29
20	21	24	35
22	23	25	5
24	25	28	14
26	27	28	32
28	29	29	31
30	31	30	4
32	33	33	4
34	35	33	14
36	37	34	17
38	39	37	26
40	41	38	35
42	43	39	36
44	45	39	28

Figure 5: Shopping Mall's Customers who have a Low Annual Income and have a Low spending score.

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Number of Customers in Each Cluster
Cluster 1: 81 customers
Cluster 2: 39 customers
Cluster 3: 22 customers
Cluster 4: 35 customers
Cluster 5: 23 customers

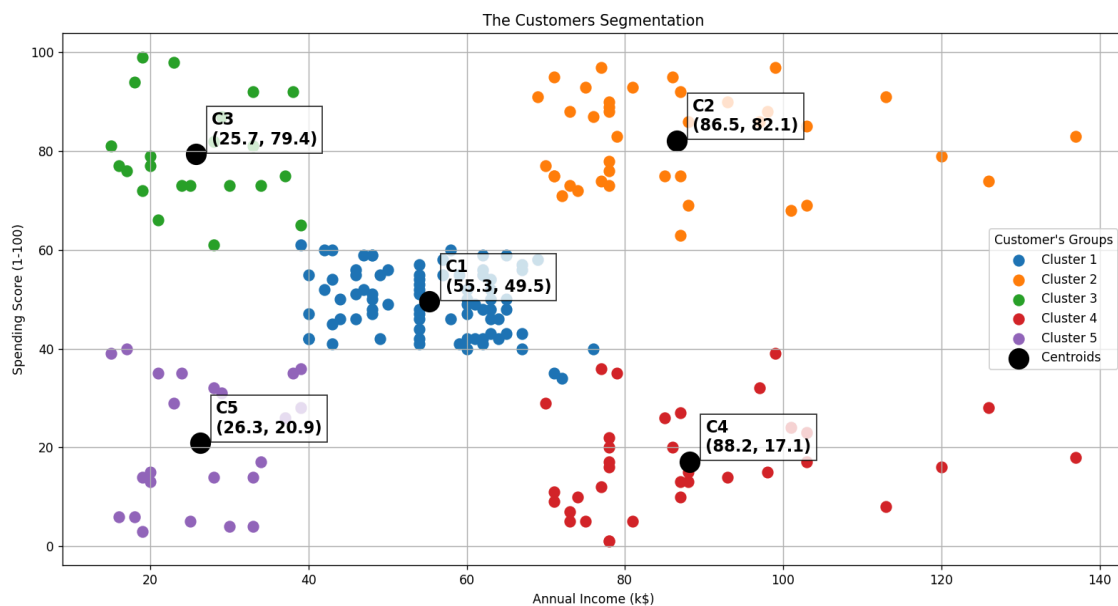
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In this image it represented the Number of Customers in Each Cluster, there are 81-Cluster 1(Average Income, Average Spending), 39-Cluster 2 (High Income, High Spending), 22-Cluster 3 (Low Income, High Spending), 35-Cluster 4 (High Income, Low Spending), 23-Cluster 5 (Low Income, Low Spending), with the result being shown it indicates that Cluster 1 has a highest number of customers, and Cluster 3 has a lowest number of customers. It indicates here that there are some customers who earn a lot but spend a little, a lot, and sometimes there are some customers who earn a little or earn just right but spend a lot. Also it indicates the customer's spending can be based somewhat to their Annual Income.

Cluster Centers

	Annual Income (k\$)	Spending Score (1-100)
0	55.296296	49.518519
1	86.538462	82.128205
2	25.727273	79.363636
3	88.200000	17.114286
4	26.304348	20.913043

And here are the cluster centers which are the centroids, those clusters of Annual Income and Spending Score with index of 0-4 (represented as 1-5) which are interpreted as the specific points that are centered on each cluster from 1-5.



In this image it is the result of a graph of K-Means of shopping malls customers based on their clusters that they belong to. And it is the graphical representation of each clusters.

Customers who are near the centroids (black dots) of cluster 1 and have a color of blue dots are belong to an Average Income and Average Spending Group.

And those customers who are near the centroids (black dots) of cluster 2 and have a color of orange dots are belong to a High Income and High Spending Group.

Meanwhile customers who are near the centroids(black dots) of cluster 3 and have a color of green dots are belong to a Low Income but High Spending Group.

And the customers who are near the centroids(black dots) of cluster 4 and have a color of red dots are belong to a High Income and High Spending Group.

And last customers who are near the centroids(black dots) of cluster 5 and have a color of purple dots are belong a to Low Income and Low Spending Group.

Conclusions:

Those results that are being presented it is clear that there are some customers who has high income but tend to spends a lot or spends a little, meanwhile there are some customers who has a low income but spends a little or spends a lot and there are some customers who has a an average income who just spend in an average level. It is concluded that spending does not always depend on a person's annual income, it can be based on their preferences and shopping malls' offers or promos that will make customers spend a lot of their money.

With the graphs that are being shown, those centroids in the graph, represented as the annual income and spending core of each cluster , serve as the center for the customer's segmentation. And the separation or clusterings clearly shows that groupings of a customer are being done well using the K-Means Clustering. In return the clustering has provided valuable insights into customer behavior which will help the business malls to develop a strategic marketing campaigns that will improve the customers satisfaction, customers buying retention and maximizing effectively the business profitability by improving the promotions and services that they will offer to their customers.

Recommendations:

These are the following recommendations that are appropriate in this report:

Cluster	Customer Type	Recommendation
Cluster 1(Centroid: 55.3, 49.5)	Average Income, Average Spending	Encourage them to purchase a lot of products by giving them loyalty programs, reward points and seasonal discounts to become high value customers.
Cluster 2(Centroid: 86.5, 82.1)	High Income, High Spending	Offered them VIP memberships, exclusive discounts and premium customer services. Because they are the one who contribute increasingly to the revenue.
Cluster 3(Centroid: 25.7, 79.4)	Low Income, High Spending	Give the customers affordable promotions and discounts like student or senior discount if possible and reward points that they can use purchasing to maintain their loyalty.
Cluster 4(Centroid: 88.2, 17.1)	High Income, Low Spending	Increased customer engagement through offering them personalized marketing, exclusive premium experiences, and limited time offers, since they are the ones who has a high purchasing ability and to encourage them to spend more.
Cluster 5(Centroid: 26.3, 20.9)	Low Income, Low Spending	Attract the customers by offering them budget friendly products, discounts, and coupons and promotional bundles to increase the customers visit frequency and occasional purchases.

Source:

- <https://www.kaggle.com/datasets/kandij/mall-customers>
- <https://github.com/nimrodzeus83-de/Custoomer-s-Segmentation-in-a-Shopping-Mall/blob/main/Mall.py>
- https://github.com/nimrodzeus83-de/Custoomer-s-Segmentation-in-a-Shopping-Mall/blob/main/Mall_Customers.csv